Notes: Coval and Moskowitz (JPE, 2001)

The Geography of Investment: Informed Trading and Asset Prices

Contents

1 Big picture

- **Question.** Do mutual fund managers have an informational advantage in geographically nearby stocks, and does this local advantage show up both in portfolio performance and in the cross section of expected stock returns?
- **Setting.** The paper studies actively managed U.S. mutual funds and compares the performance of their *local* holdings—stocks headquartered within 100 kilometers of the fund manager—to their distant holdings and to nearby stocks they choose not to hold.
- **Core idea.** Geography can be used as an empirical lens to isolate the part of a manager's portfolio where informational advantages should be strongest. If local information matters, then nearby holdings should outperform other holdings after controlling for risk.
- **Main empirical result.** The average fund manager earns an additional raw return of **2.67 percentage points per year** on local holdings relative to distant holdings. After adjusting for size, book-to-market, and momentum, local holdings still outperform distant holdings by **1.18 percentage points per year**. In addition, local stocks actually held outperform local stocks ignored by the manager by about **3 percentage points per year**, and local buys outperform local sells by about **1.24 percentage points per year**.
- **Who has the strongest local advantage?** The local premium is strongest among funds that are **small**, **old**, concentrated in **few holdings**, and located in **remote areas**. These are exactly the kinds of managers who should be best able to exploit local information.
- **Stock-level result.** Firms held disproportionately by nearby mutual fund managers earn higher subsequent returns. A zero-cost strategy that buys the most locally owned stocks and shorts the least locally owned stocks earns about **2.56 percentage points per year raw** and roughly **1.1 percentage points risk-adjusted**. The effect is much stronger among small firms.
- **Economic interpretation.** Local investors appear to trade local securities at an informational advantage. This provides rare evidence that some mutual fund managers *do* select stocks successfully, but only in the geographic segment of their portfolios where their information edge is most likely to exist.
- **Takeaway.** Geography matters for information, portfolio choice, and asset prices. The paper therefore links three ideas that are usually studied separately: local bias, informed trading, and expected returns.

2 Data and measurement (key objects)

2.1 Observed fund-level and stock-level data

- The core fund-holdings data come from the **CDA Investment Technologies** common stock holdings and transactions tapes.
- These data report the quarterly equity holdings of virtually all U.S. mutual funds from **January 1975 to December 1994**.

- Fund locations are obtained by matching each fund's management company to **Nelson's Directory of Investment Managers** and then translating city and state into latitude and longitude using the **Geographic Names Information System Digital Gazetteer (GNISDG)**.
- Stock returns come from **CRSP**. Firm headquarters locations come from **Disclosure**, again translated into latitude and longitude coordinates through the GNISDG.
- The analysis excludes index funds and focuses on active managers. It also excludes funds with fewer than five equity holdings.

2.2 Sample construction

- After restrictions, the number of funds in the sample falls from **393 to 150** in 1975 and from about **2,400 to 1,258** in 1994.
- The mutual funds in the sample hold between **330 and 4,617** different companies over time, and the average fund holds equity in roughly **40** firms.
- Only CRSP-listed domestic equities are used for return measurement. Portfolio weights are recomputed so that CRSP-listed equities sum to one within each fund.
- The paper uses the firm's **headquarters location**, not its state of incorporation, because state of incorporation is often chosen for legal reasons and need not correspond to the location of actual operations.

2.3 Markets, funds, and local investment

- A stock is defined as **local** to fund i if the headquarters of the firm is within **100 kilometers** of the fund's headquarters.
- The paper also checks an alternative fund-specific definition of localness based on being 95% closer than the average stock in the market; this gives qualitatively similar results.
- The relevant "market" for computing a fund's local bias is the set of investable stocks held by at least one fund in the sample.

2.4 Key descriptive objects

- The average fund allocates **6.95%** of assets to local stocks, while the fraction of the market portfolio available within 100 kilometers is only **6.16%**. Thus the average local bias is approximately **0.79 percentage points**.
- This average local tilt is statistically strong but economically modest.
- The local share is not uniform across funds. Some funds hold very little locally, while highly locally biased funds tilt as much as **20–25%** of assets toward nearby stocks.
- This heterogeneity is crucial for the paper because the strongest local performance comes from the subset of managers who actually choose to concentrate locally.

2.5 Performance objects

The paper studies three related local-performance objects:

- the performance of **local holdings** relative to **distant holdings**;
- the performance of **local stocks held** relative to **local stocks not held**;
- the performance of **local buys** relative to **local sells**.

Interpretation. These three comparisons progressively strengthen the information story. Local-versus-distant could partly reflect different stock characteristics; local-held versus local-not-held gets closer to manager selection; and local-buys versus local-sells isolates the information content of trades.

3 Models and empirical specifications (all equations + interpretation)

3.1 Local versus distant portfolio returns

Each quarter, a fund is divided into a local and a distant portfolio. Using the reported holdings from the *previous* quarter, the paper computes value-weighted excess returns over the following three months:

$$\widetilde{R}^{L}_{i,t} = \frac{1}{3}\sum_{z=1}^{3}\sum_{j=1}^{L_{i,t}} w^{L}_{ij,t}\,\widetilde{r}_{j,t+z}, \qquad \widetilde{R}^{D}_{i,t} = \frac{1}{3}\sum_{z=1}^{3}\sum_{j=1}^{D_{i,t}} w^{D}_{ij,t}\,\widetilde{r}_{j,t+z},$$

where:

- $\widetilde{r}_{j,t}$ is stock j's return in excess of the three-month Treasury bill rate,
- $w^{L}_{ij,t}$ and $w^{D}_{ij,t}$ are portfolio weights within the local and distant subportfolios,
- $L_{i,t}$ and $D_{i,t}$ denote the number of local and distant firms held by fund i.

Interpretation. The previous-quarter-holdings convention is conservative. It avoids contamination from reporting lags and from "window dressing," where managers buy recent winners just before disclosure dates.

3.2 Risk adjustment

The paper adjusts stock returns using the **Daniel et al. (1997)** characteristic-based benchmark procedure.

- All stocks are sorted into size quintiles.
- Within size quintiles, they are sorted into book-to-market quintiles.
- Within those groups, they are sorted into momentum quintiles.

This creates **125 benchmark portfolios**. A stock's matched-portfolio return is subtracted from its raw return.

Interpretation. The risk adjustment controls for the main empirical dimensions of average return differences: size, value, and momentum. So a remaining local premium is not just a mechanical small-stock or value-stock effect.

3.3 Local stocks not held by local funds

The paper compares the performance of local stocks actually held by a nearby manager with the performance of other local stocks the manager *does not* hold.

Interpretation. This is a powerful test against a simple neglected-stock story. If all nearby stocks were just underfollowed or locally neglected, then *all* local stocks should do well. Instead, only the local stocks the manager chooses to hold perform unusually well.

3.4 Local buy versus local sell portfolios

To study whether trades reveal information, the paper forms two local portfolios:

- a portfolio of local stocks in which the fund **increased** its position,
- a portfolio of local stocks in which the fund **decreased** its position.

The return difference between these two portfolios measures whether local purchases outperform local sales.

Interpretation. If managers truly possess local information, then changes in local holdings should forecast future returns even more sharply than levels of holdings.

3.5 Persistence of local performance

The paper reconstructs the three local-performance strategies using additional lags of **1, 3, 6, and 12 months** between the reported quarterly holdings and realized returns.

Interpretation. This is a direct test of whether outsiders could mimic informed local trades using publicly available SEC filings. If local abnormal returns disappear quickly, then outsiders cannot easily free-ride on local information.

3.6 Fund local bias measure

For each fund, local bias is defined as

$$\text{Local Bias}_i = \frac{\text{fund assets in local stocks}}{\text{fund assets}} - \frac{\text{market capitalization of local stocks}}{\text{investable market capitalization}}.$$

Interpretation. This corrects for the fact that funds in New York or Boston naturally have more firms nearby than funds in remote cities. A manager is locally biased only if she holds more local stock than the local market opportunity set would imply.

3.7 Stock-level local ownership measure

For stock j, local ownership is defined as

$$LO_j = \frac{\sum_{i \in \mathcal{N}(j)} D_{ij}}{\sum_{i=1}^{M} D_{ij}} - \frac{\sum_{i \in \mathcal{N}(j)} V_i}{\sum_{i=1}^{M} V_i},$$

where:

- $\mathcal{N}(j)$ is the set of mutual funds located within 100 kilometers of firm j's headquarters,
- D_{ij} is fund i's dollar holding of stock j,
- V_i is fund i's total asset value,
- M is the total number of funds.

Interpretation. The first term measures how much of stock j is held by nearby funds; the second term subtracts the baseline share of mutual fund capital that simply happens to be located near firm j. Thus $LO_j > 0$ means the stock is disproportionately owned by local funds.

3.8 Fama–MacBeth cross-sectional return regressions

To study whether local ownership predicts returns, the paper runs monthly cross-sectional regressions of the form

$$R_{j,t} = \alpha_t + \beta_t LO_{j,t-1} + \Gamma_t X_{j,t-1} + \varepsilon_{j,t},$$

where $X_{j,t-1}$ includes:

- log market capitalization,
- book-to-market equity,
- past one-month return,
- past 12- to 2-month return,
- past 36- to 13-month return,
- past industry return,
- past regional return,
- a remote-city indicator.

The coefficients are averaged over time in the style of **Fama and MacBeth (1973)**.

Interpretation. This asks whether stocks overweighted by nearby mutual fund managers earn higher average returns, controlling for the classic return predictors already emphasized in asset pricing.

3.9 Abnormal turnover and herding

For fund i holding stock j, abnormal turnover is defined as

$$ab\,t_{ij,t} = \frac{|H_{ij,t} - H_{ij,t-1}|}{H_{ij,t-1}} - \mathbb{E}\left[\frac{|H_{ij,t} - H_{ij,t-1}|}{H_{ij,t-1}}\right],$$

where $H_{ij,t}$ is the number of shares of stock j held by fund i.

The stock-level herding measure is

$$HM_{j,t} = \left| \frac{\#buyers_{j,t}}{\#traders_{j,t}} - \frac{\#buys_t}{\#trades_t} \right|.$$

Interpretation. The turnover measure asks whether local positions are traded more or less aggressively than distant ones. The herding measure asks whether local investors follow the crowd less than distant investors. If local investors are better informed, they should trade local stocks more selectively and herd less.

4 Identification and interpretation (what makes the empirical case credible)

4.1 Why geography helps identify information advantage

- The paper does *not* claim that mutual funds outperform in general.
- Instead, it compares the *local segment* of a manager's portfolio to the *distant segment*. This is an internal comparison within the same class of investor.
- The idea is that geographic proximity should matter exactly where information is costly, tacit, and difficult to transmit: visiting plants, talking to suppliers, understanding local demand conditions, and potentially gaining access to soft information through local social networks.

Interpretation. Geography is not the information itself; it is an empirical proxy for where information acquisition should be easiest.

4.2 Why the local-not-held comparison is important

A simple alternative story is that local firms are just neglected, illiquid, or otherwise mispriced for reasons unrelated to manager skill.

- If that were true, then *all* nearby firms should earn high subsequent returns.
- Instead, local firms *not held* by local managers underperform the local firms the managers do hold.

Interpretation. This sharpens the evidence from "local investing is good" to "local stock *selection* is good." That is a much stronger statement.

4.3 Why fund heterogeneity matters

The local-information story predicts stronger effects for funds that can actually exploit soft information.

- **Small funds** are more agile and can take concentrated positions in small local firms.

- **Funds with few holdings** can devote more attention to each name.
- **Older funds** may have deeper local ties and lower career-concern costs from deviating from the market portfolio.
- **Remote-city funds** may face less local competition from other informed money managers.

Interpretation. The fact that the empirical results line up with these predictions gives the information interpretation much more credibility than a purely mechanical local-bias story.

4.4 Why the stock-level evidence matters

The stock-level return regressions and the zero-cost local-ownership portfolios matter because they move from the *manager* perspective to the *stock* perspective.

- If local managers are informed, then firms disproportionately held by nearby funds should command higher future expected returns.
- That is exactly what the paper finds, especially among small stocks and among stocks heavily held by mutual funds.

Interpretation. This links local information to asset prices, not just to fund performance.

4.5 Limits and caveats

- The paper cannot cleanly separate all sources of local information. It could be monitoring, soft information, private information, or a combination.
- The analysis uses fund headquarters and firm headquarters. That is a practical measure of proximity, but actual information networks could be more complicated.
- The persistence results suggest that outsiders might replicate some local gains if disclosure is timely, but the paper does not incorporate realistic transaction costs in those mimic strategies.
- The local performance premium is strongest in the first half of the sample when comparing local to distant holdings, so the magnitude of the effect is not literally constant over time.

4.6 Relation to the literature

- Relative to the mutual fund performance literature, the paper provides rare evidence of positive stock-picking skill, but only in a particular slice of the portfolio.
- Relative to the home-bias and geography literature, it adds a key missing dimension: local investment is not just a preference or familiarity phenomenon; it is also associated with better performance.
- Relative to the asymmetric-information asset-pricing literature, the paper offers an empirical way to identify a set of investors who plausibly possess superior information.

5 Tables and figures

5.1 Table 1: local bias and local performance

Read:

- The average fund allocates **6.95%** to local stocks, versus **6.16%** in the nearby market portfolio, implying an average local bias of about **0.79 percentage points**.
- Raw excess returns on local holdings are **8.71%** per year versus **6.04%** on distant holdings, a difference of **2.67%**.
- After characteristic adjustment, local holdings earn **1.84%** versus **0.66%** for distant holdings, so the risk-adjusted local premium is **1.18%** per year.
- Local stocks actually held outperform local stocks not held by about **3.01%** per year on a risk-adjusted basis.
- Local buys outperform local sells by about **1.24%** per year.
- The comparison of subperiods shows that the local-versus-distant premium is much stronger in 1975–84 than in 1985–94, but the local-held versus local-not-held and local-buy versus local-sell evidence remains much more stable across subperiods.

TABLE 1

PERFORMANCE OF LOCAL FUND POSITIONS (Percentage Annualized Returns)

A. LOCAL BIAS AND RAW RETURNS OF LOCAL VS. DISTANT HOLDINGS

	ASSETS RESIDING LOCALLY			RAW EXCESS RETURNS		
	% Held ≤100 km (1)	% Market ≤100 km (2)	Difference (3)	Local: $\tilde{R}^L$ (4)	Distant: $\tilde{R}^D$ (5)	$\tilde{R}^L - \tilde{R}^D$ (6)
1975–94	6.95	6.16	.79	8.71	6.04	2.67
			(10.60)	[.36]	[.42]	(3.26)
1975–84	6.73	5.85	.89	12.07	6.57	5.50
			(7.03)	[.46]	[.46]	(4.15)
1985–94	7.18	6.48	.70	5.52	5.36	.16
			(8.62)	[.38]	[.24]	(.17)

B. RISK-ADJUSTED RETURNS OF LOCAL FUND HOLDINGS

	Local: $\tilde{R}^L$ (1)	Distant: $\tilde{R}^D$ (2)	$\tilde{R}^L - \tilde{R}^D$ (3)	Local, Not Held: $\tilde{R}^{LN}$ (4)	$\tilde{R}^L - \tilde{R}^{LN}$ (5)	$\tilde{R}^{L^+} - \tilde{R}^{L^-}$ (6)
1975–94	1.84	.66	1.18	−1.17	3.01	1.24
	[.33]	[.22]	(3.49)	[−.29]	(7.55)	(2.87)
1975–84	2.32	.00	2.32	−2.05	4.37	1.27
	[.42]	[.00]	(4.84)	[−.42]	(5.80)	(2.36)
1985–94	1.36	1.32	.04	−.68	2.04	1.20
	[.24]	[.38]	(.09)	[−.10]	(4.82)	(2.64)

NOTE.—Every quarter from January 1975 to December 1994, each fund is split into a "local" portion (defined as any holding located within 100 kilometers of the fund manager's location) and a "distant" portion. The dollar-weighted average annualized return and Sharpe ratio (in brackets) of these portfolios are computed for each fund every month and then averaged across all funds (value-weighted by total asset value) and reported, along with the average difference between them (t-statistics are in parentheses). Panel A reports the raw returns in excess of the three-month Treasury bill rate for the local and distant fund portions and reports the fraction of fund assets devoted to local equities (i.e., within 100 kilometers), the fraction of total market capitalization that exists within 100 kilometers of the fund, and their difference. Panel B reports the risk-adjusted returns (adjusted for size, BE/ME, and momentum via Daniel et al. [1997]) of the local and distant fund portions, the risk-adjusted return of local stocks *not being held* by local funds ($\tilde{R}^{LN}$), the difference between the performance of these firms and the local stocks actually being held, and the risk-adjusted return difference between the local buy and local sell portfolios, defined as the portfolio of local stocks held by funds that had an increase in shares ($\tilde{R}^{L^+}$) and a decrease in shares ($\tilde{R}^{L^-}$), respectively. Statistics are reported over the whole sample period and for each half of the sample separately. Reported returns are expressed in annual percentage rates.

TABLE 2

REPORTING LAGS AND LOCAL FUND PERFORMANCE (Percentage Annualized Returns)

	LAGS BETWEEN REPORTED HOLDINGS AND RISK-ADJUSTED RETURNS			
	1 Month	3 Months	6 Months	12 Months
$\tilde{R}^L - \tilde{R}^D$	1.06	1.01	.01	−.50
	(3.13)	(2.41)	(.04)	(−1.94)
$\tilde{R}^L - \tilde{R}^{LN}$	3.05	2.86	1.51	.60
	(7.55)	(6.23)	(2.22)	(.91)
$\tilde{R}^{L^+} - \tilde{R}^{L^-}$	1.01	.90	.13	−.20
	(2.11)	(1.88)	(.15)	(−.54)

NOTE.—Risk-adjusted returns of the difference between local and distant fund portions, the difference between local stocks being held and local stocks not being held by funds, and the difference between the local buy and local sell portfolios are reported using various lags between the reported fund holdings and returns over the January 1975 to December 1994 sample period. t-statistics are in parentheses. Reported returns are expressed in annual percentage rates.

5.2 Table 2: persistence and the possibility of imitation

Read:

- The risk-adjusted local-minus-distant strategy still earns about **1.06%** with a one-month lag and **1.01%** with a three-month lag, but it disappears by six months.

- The local-held minus local-not-held strategy is even more persistent: about **3.05%** at one month, **2.86%** at three months, and **1.51%** at six months, before fading by 12 months.
- The local-buy minus local-sell premium is strongest only at short lags, consistent with trades containing shorter-horizon information than holdings levels.
- Economically, the table suggests that if SEC holdings disclosures become available quickly enough, outside investors might replicate some local-information profits. But this is only a gross-return statement; it may disappear net of costs.

5.3 Table 3: which funds concentrate locally?

Read:

- Only the top local-bias funds appear to earn strong local abnormal returns.
- In the top quintile of local bias, funds place about **22.54%** of assets in local stocks compared with a local market share of **8.67%**, for a bias of **13.86 percentage points**.
- These highly local funds earn local-minus-distant abnormal returns of about **1.28%** and local-held minus local-not-held abnormal returns of about **4.14%**.
- Quintile 4 also performs strongly: local-minus-distant returns of about **3.18%** and local-held minus local-not-held of about **5.41%**.
- By contrast, the first three quintiles of funds do not show convincing local skill. This is important because it means the paper is not just documenting that *all* mutual funds benefit mechanically from local investing.

TABLE 3
Local Fund Performance for Various Degrees of Local Bias (Percentage Annualized Returns)

	Fraction of Fund Local Ownership Quintiles							
	Assets Residing Locally			Risk-Adjusted Returns				
Quintile	% Held	%Market	Difference	$\tilde{R}^L$	$\tilde{R}^D$	$\tilde{R}^L - \tilde{R}^D$	$\tilde{R}^{LN}$	$\tilde{R}^L - \tilde{R}^{LN}$
Q1	1.69	10.47	−8.78	1.07	.98	.09	−1.23	2.29
			(−64.74)	[.15]	[.17]	(.02)	[−.27]	(3.99)
Q2	1.33	4.09	−2.76	−.28	1.23	−1.50	−.01	−.27
			(−65.48)	[−.06]	[.25]	(−3.53)	[−.00]	(−.57)
Q3	1.83	3.13	−1.30	.92	.65	.27	.11	.81
			(−56.10)	[.13]	[.14]	(.53)	[.02]	(1.35)
Q4	5.81	4.62	1.19	3.54	.36	3.18	−1.87	5.41
			(14.71)	[.40]	[.10]	(5.71)	[−.27]	(8.49)
Q5	22.54	8.67	13.86	1.66	.37	1.28	−2.48	4.14
			(53.81)	[.19]	[.09]	(2.27)	[−.26]	(5.26)
Q5−Q1	20.85	−1.79	22.64	.59	−.61	1.19	−1.26	1.85
			(77.98)	[.05]	[−.08]	(2.16)	[−.14]	(1.86)

Note.—The local performance measures of active mutual funds are reported over the January 1975 to December 1994 time period for various funds exhibiting different degrees of local bias. The table reports the local and distant risk-adjusted fund performance measures for quintiles of funds sorted by their fraction of local ownership in excess of the market. This measure is defined as the fraction of fund assets devoted to stocks within 100 kilometers of the fund minus the fraction of total market capitalization that exists within 100 kilometers of the fund. We report these measures along with their difference for comparison across the quintiles. Also reported are the risk-adjusted returns of local stocks *not being held* by local funds as well as the difference between local stocks not held and local stocks actually held. *t*-statistics are in parentheses and Sharpe ratios are in brackets. Reported returns are expressed in annual percentage rates.

TABLE 4
Local Fund Performance across Other Fund Attributes (Percentage Annualized Returns)

	Assets Residing Locally			Risk-Adjusted Returns				
	% Held	% Market	Difference	$\tilde{R}^L$	$\tilde{R}^D$	$\tilde{R}^L - \tilde{R}^D$	$\tilde{R}^{LN}$	$\tilde{R}^L - \tilde{R}^{LN}$
	Fund Size (Total Asset Value)							
Q5	6.82	6.34	.48	.24	.85	−.61	−.15	.39
(highest)			(3.63)	[.07]	[.26]	(−2.35)	[−.03]	(1.20)
Q1	6.90	6.00	.90	1.99	.59	1.41	−1.29	3.28
(lowest)			(9.03)	[.31]	[.18]	(3.46)	[−.29]	(7.09)
	Number of Holdings							
Q5	5.86	6.57	−.71	−.30	.56	−.86	−.89	.59
			(−2.97)	[−.07]	[.10]	(−2.03)	[−.17]	(1.43)
Q1	7.37	6.06	1.31	3.08	.92	2.16	−1.64	4.72
			(11.33)	[.32]	[.22]	(3.48)	[−.19]	(6.11)
	Age of Fund							
Q5	7.69	5.93	1.76	2.20	.82	1.38	−1.03	3.22
			(16.51)	[.31]	[.26]	(3.21)	[−.23]	(6.24)
Q1	6.64	6.93	−.29	1.35	.54	.81	−.77	2.12
			(−1.63)	[.24]	[.14]	(1.81)	[−.13]	(4.74)
	Metropolitan Area							
Remote	5.91	4.25	1.67	2.28	1.04	1.25	−1.09	3.37
			(15.00)	[.26]	[.23]	(2.24)	[−.18]	(5.29)
Small	7.33	6.56	.77	1.95	.82	1.12	−1.10	3.05
			(7.58)	[.31]	[.24]	(2.69)	[−.23]	(6.84)
Large	10.14	9.24	.90	1.24	.28	.97	−1.03	2.28
			(5.57)	[.21]	[.07]	(2.12)	[−.18]	(4.91)

Note.—The local performance measures of active mutual funds are reported over the January 1975 to December 1994 time period for various types of funds. Risk-adjusted returns are reported for the top (Q5) and bottom (Q1) quintiles of funds on the basis of several fund attributes: size (total asset value), number of holdings, and age. In addition, we report returns for funds located in large, small, and remote cities. Also reported are the fraction of fund assets devoted to stocks within 100 kilometers and the fraction of total market capitalization that exists within 100 kilometers of the fund. We report these measures along with their difference for comparison across fund attributes. Finally, the risk-adjusted returns of local stocks *not being held* by local funds as well as the difference between local stocks not being held and local stocks actually held are reported for each fund group. *t*-statistics are in parentheses and Sharpe ratios are in brackets. Reported returns are expressed in annual percentage rates.

5.4 Table 4: what kind of manager has local skill?

Read:

- **Small funds** outperform: the smallest quintile earns local-minus-distant abnormal returns of about **1.41%** and local-held minus local-not-held of about **3.28%**; the largest quintile shows much weaker local performance.
- **Focused funds** outperform: funds with the fewest holdings earn local-minus-distant returns of about **2.16%** and held-versus-not-held returns of about **4.72%**; widely diversified funds show essentially no local skill.
- **Older funds** outperform: the oldest quintile earns local-minus-distant returns of about **1.38%** and held-versus-not-held returns of about **3.22%**; the youngest funds show much weaker effects.
- **Remote funds** have the strongest local advantage: remote-city funds show held-versus-not-held abnormal performance of about **3.37%** and a local bias of **1.67 percentage points**; large-city funds exhibit smaller local advantages.

Interpretation. This table is one of the strongest pieces of evidence for the information story because the sign and ranking of the results line up with economic intuition about who should possess and exploit local knowledge.

5.5 Table 5: local ownership and the cross section of expected returns

Read:

- In Fama–MacBeth regressions, the coefficient on local ownership is positive and significant in the full sample and particularly strong for **small stocks**.
- The local-ownership coefficient is about **0.0016** in the full sample, **0.0036** in the smallest-size quintile, and essentially zero for the largest stocks.
- The usual cross-sectional return patterns still remain: small firms, high book-to-market firms, and momentum variables matter in the expected way.
- The local-ownership variable does *not* simply absorb those other characteristics. It adds independent explanatory power.

5.6 Table 6: local ownership sorted portfolios

Read:

- A zero-cost strategy that buys the highest-local-ownership quintile and shorts the lowest earns **2.56%** raw per year.
- After Daniel et al. risk adjustment, the premium is still about **1.10%**; using a Carhart four-factor alpha, it is about **1.12%**.
- The premium is much stronger among **small firms**: **4.18%** raw, **3.00%** characteristic-adjusted, and **3.56%** using four-factor alpha.

TABLE 5
IMPACT OF LOCAL INVESTMENT ON THE CROSS SECTION OF EXPECTED RETURNS
Dependent Variable: Cross Section of Stock Returns

	All Stocks Held by Funds		Smallest-Size Quintile		Largest-Size Quintile	
Log(size)	−.0004	−.0009	.0017	−.0013	−.0008	−.0012
	(−1.94)	(−2.78)	(5.15)	(−1.57)	(−1.15)	(−1.67)
BE/ME	.0019	.0025	−.0030	.0012	.0009	.0006
	(2.13)	(2.47)	(−2.02)	(.66)	(.81)	(.45)
$ret_{-1:-1}$	−.0595	−.0599	−.0699	−.0581	−.0390	−.0386
	(−10.57)	(−8.47)	(−8.51)	(−5.43)	(−3.90)	(−3.40)
$ret_{-12:-2}$	.0081	.0090	.0047	.0067	.0018	.0026
	(4.19)	(3.78)	(1.86)	(2.16)	(.59)	(.77)
$ret_{-13:-36}$	−.0013	−.0014	−.0031	−.0046	.0012	.0009
	(−1.47)	(−1.46)	(−1.40)	(−1.88)	(.70)	(.51)
$ind_{-12:-1}$	.0130	.0094	.0135	.0029	.0149	.0142
	(2.35)	(1.55)	(1.36)	(.30)	(2.14)	(1.75)
$region_{-12:-1}$	.0076	.0177	.0028	−.0006	.0360	.0320
	(.89)	(1.72)	(.12)	(−.02)	(2.03)	(1.50)
Remote city	−.0007	.0000	.0033	.0004	−.0004	−.0016
	(−1.26)	(.01)	(1.22)	(.22)	(−.43)	(−1.60)
LO		.0016		.0036		.0006
		(2.78)		(3.30)		(.18)

NOTE.—Cross-sectional regressions of stock returns on local ownership and a host of firm characteristics are run every month on the universe of individual stocks held by at least one mutual fund in our sample over the period January 1975 to December 1994. The cross sections of stock returns at time t are regressed on a constant (not reported), log of size (market capitalization at $t-1$), BE/ME, several individual and industry past return variables, the past-year return of the region in which each stock is headquartered, a remote city dummy (indicating whether the stock's headquarters is outside of 250 kilometers of any of the 20 largest populated cities), and the local ownership variable. The local ownership measure, LO, is the difference between the percentage of mutual fund dollars devoted to stock j that are provided by local mutual fund managers (within 100 kilometers) and the percentage of total mutual fund assets that resides within 100 kilometers of stock j's headquarters. These regressions are repeated for the smallest and largest quintiles of stocks. The time-series average of the coefficient estimates and their associated time-series t-statistics (in parentheses) are reported in the style of Fama and MacBeth (1973).

TABLE 6
LOCAL OWNERSHIP SORTED PORTFOLIOS (Percentage Annualized Returns)

	Raw Returns			Risk-Adjusted Returns: Daniel et al.	Risk-Adjusted Returns: Four-Factor α
	LO Q1	LO Q5	Q5−Q1	Q5−Q1	Q5−Q1
Unconditional	7.03	9.59	2.56	1.10	1.12
	(3.43)	(4.52)	(2.24)	(1.74)	(1.86)
	Size (Market Capitalization)				
Small	13.62	17.80	4.18	3.00	3.56
	(3.77)	(4.59)	(2.22)	(2.18)	(3.23)
Medium	10.57	12.86	2.29	−1.14	1.90
	(3.46)	(4.05)	(1.74)	(−1.05)	(1.32)
Large	8.22	8.25	.03	−.58	−.61
	(3.56)	(4.06)	(.09)	(−1.09)	(−1.11)
	Fraction of Equity Held by Mutual Funds				
Low	12.20	11.63	−.57	.19	−.81
	(3.74)	(3.95)	(−.25)	(.10)	(−.33)
Middle	8.98	8.52	−.47	.39	1.21
	(3.91)	(3.71)	(−.29)	(.30)	(.70)
High	4.29	8.98	4.68	2.68	3.09
	(2.14)	(3.88)	(2.90)	(2.35)	(1.81)
	Book-to-Market Equity				
Low	3.54	7.69	4.16	1.08	1.93
	(1.38)	(2.83)	(2.58)	(.81)	(.81)
Middle	8.59	9.78	1.19	.56	1.44
	(3.34)	(3.88)	(.67)	(.37)	(.74)
High	11.01	12.28	1.27	−.12	−1.72
	(4.73)	(4.64)	(.72)	(−.08)	(−.90)
	Book-to-Market Equity, Smallest Half of Firms				
Low	1.53	7.05	5.52	1.57	2.47
	(.60)	(2.66)	(3.41)	(1.14)	(1.58)
Middle	7.25	6.06	−1.19	2.76	2.89
	(3.03)	(2.45)	(−.65)	(1.68)	(1.72)
High	10.73	12.70	1.97	−.18	−.25
	(4.85)	(5.22)	(1.19)	(−.14)	(−.36)

NOTE.—Every month stocks are sorted into quintiles on the basis of LO from the previous quarter, which is the difference between the percentage of mutual fund dollars devoted to stock j that are provided by local mutual fund managers (within 100 kilometers) and the percentage of total mutual fund assets that resides within 100 kilometers of stock j's headquarters. Value-weighted returns of the quintile portfolios are computed every month, and time-series averages and t-statistics (in parentheses) are reported for the highest- and lowest-LO quintile portfolios (LO Q5 and LO Q1, respectively), as well as the difference between them (Q5−Q1) over the period January 1975 to December 1994. Also reported is the risk-adjusted return of the highest- minus lowest-LO quintile portfolio (Q5−Q1) using both the Daniel et al. (1997) characteristic adjustment and Jensen's α from a regression of this portfolio return on the Carhart (1997) four-factor model. These returns are also reported for double sorts on size (market capitalization), fraction of outstanding shares held by mutual funds, and BE/ME, where stocks are first sorted into size (mutual fund ownership or BE/ME) tretiles and then sorted within each tretile into quintiles on the basis of LO. The BE/ME sort is also repeated for only the smallest half of firms. Returns are reported as annual percentages with time-series t-statistics in parentheses.

TABLE 7
TRADING ACTIVITY AND LOCAL OWNERSHIP

	ABNORMAL TURNOVER IN HOLDINGS $\tau^{ab}_{ij,t}$			HERDING IN STOCKS: $HM_{j,t}$
	Only Fund Attributes (1)	Only Stock Characteristics (2)	Fund Attributes and Stock Characteristics (3)	(4)
Local ownership	−.2899	−1.1930	−1.2670	−.0183
	(−2.50)	(−3.22)	(−3.35)	(−4.98)
Log(fund assets)	.0001		.0001	
	(3.93)		(3.57)	
Log(#holdings)	.0073		.0077	
	(2.59)		(2.62)	
Age	.0168		.0143	
	(2.04)		(2.15)	
Remote city (fund)	.0778		.0685	
	(2.72)		(2.56)	
Fund $ret_{-36:-1}$	.1822		.3059	
	(1.84)		(3.56)	
Log(size)		.0839	.0831	−.0083
		(9.23)	(8.93)	(−11.75)
BE/ME		.0377	.0450	.0001
		(1.15)	(1.30)	(.10)
$ret_{-1:-1}$		.1383	.1334	−.0255
		(.98)	(.96)	(−2.80)
$ret_{-12:-2}$		−.0819	−.1027	−.0061
		(−1.63)	(−2.03)	(−1.88)
$ret_{-36:-13}$		.0447	.0433	.0010
		(1.50)	(1.39)	(1.03)
$ind_{-12:-1}$		.0451	.0264	.0166
		(.38)	(.23)	(1.93)
$region_{-12:-1}$		1.1410	1.0820	.0073
		(5.78)	(5.60)	(.30)
IPO		.4466	.3988	−.0050
		(6.88)	(7.26)	(−.95)
NASDAQ		−.4598	−.4634	.0188
		(−11.72)	(−11.86)	(7.39)
Standard deviation		−.2155	−.3907	.2480
		(−.67)	(−1.29)	(6.84)
Remote city (stock)		.04269	.0382	−.0060
		(1.51)	(1.43)	(−4.62)

NOTE.—Cross-sectional regressions of fund trading activity on a host of fund and stock attributes plus the local ownership variable are reported over the period January 1975 to December 1994. The dependent variable in cols. 1–3 is the abnormal turnover of each holding, $\tau^{ab}_{ij,t}$. For fund i holding firm j, this is defined as the number of shares bought or sold in firm j divided by the number of shares held in the previous quarter by fund i in firm j, minus the *expected* value of turnover in firm j. The expected value of turnover in firm j is approximated by the average daily trading volume in firm j over the prior year. The dependent variable in col. 4 is the cross section of firm herding measures, $HM_{j,t}$, defined for stock j as the absolute value of the proportion of funds trading stock j that are buyers minus the proportion of total buys to total trades in all stocks across all funds at time t. The time-series averages of the coefficient estimates are reported along with their time-series t-statistics (in parentheses) in the style of Fama and MacBeth (1973). The constant is not reported for brevity.

- The premium is also strong among firms with a **high mutual fund ownership share**: about **4.68%** raw and **2.68–3.09%** risk-adjusted.
- Among large firms there is essentially no local-ownership spread, which fits the idea that information should matter most where firms are less followed and mutual funds account for a more meaningful share of ownership.

5.7 Table 7: turnover and herding

Read:

- Abnormal turnover is **negatively related** to local ownership once fund and firm characteristics are controlled for. Managers trade distant positions more than local positions.
- This suggests that managers use more persistent, conviction-based strategies in local firms and more short-term, volatile views in distant firms.
- Herding is also **negatively related** to local ownership. Managers herd less in nearby firms and herd more in distant firms.
- The combination of low turnover and low herding in local stocks is exactly what one would expect if local investors possess superior private or soft information and therefore do not need to follow the crowd.

6 Short summary

- **Method.** The paper uses geography to decompose mutual fund portfolios into local and distant positions and then studies fund-level performance, stock-level expected returns, and trading behavior.
- **Core evidence.** Local holdings outperform distant holdings after risk adjustment; local stocks held outperform local stocks ignored; and local buys outperform local sells.
- **Who exploits the advantage?** The strongest local skill belongs to small, old, focused, and remote funds—precisely the managers who should be best placed to gather and use local information.
- **Asset-pricing implication.** Firms disproportionately owned by nearby funds earn higher subsequent returns, especially small firms and firms heavily owned by mutual funds.
- **Trading implication.** Local investors trade local stocks less frequently and herd less, which is consistent with an information advantage rather than mechanical local preference.

7 Conclusion

7.1 Core conclusion

- The paper provides rare evidence of positive mutual fund stock-picking ability.
- That ability is not visible in the manager's overall portfolio, but it is clearly visible in the geographically proximate slice of the portfolio.
- The stock-level evidence further suggests that local information is sufficiently valuable to affect expected returns.

7.2 Economic intuition

- Proximity lowers the cost of acquiring soft information.
- Nearby managers can monitor firms more easily, understand local operating conditions better, and perhaps access private information through local networks.
- Because these advantages are not equally available to distant investors, nearby holdings earn abnormal returns.
- At the same time, only a subset of managers exploit this advantage strongly, which is why the average mutual fund still does not look impressive in aggregate performance studies.

7.3 Takeaway

This paper links geography to information and then information to asset prices. It shows that local bias is not merely a behavioral curiosity or a familiarity preference. In the mutual fund setting, local investment appears to be a rational response to information frictions. The deeper lesson is that one can often discover informed trading not by looking at *who* performs well overall, but by identifying the precise segment of the market where their informational edge should be strongest.